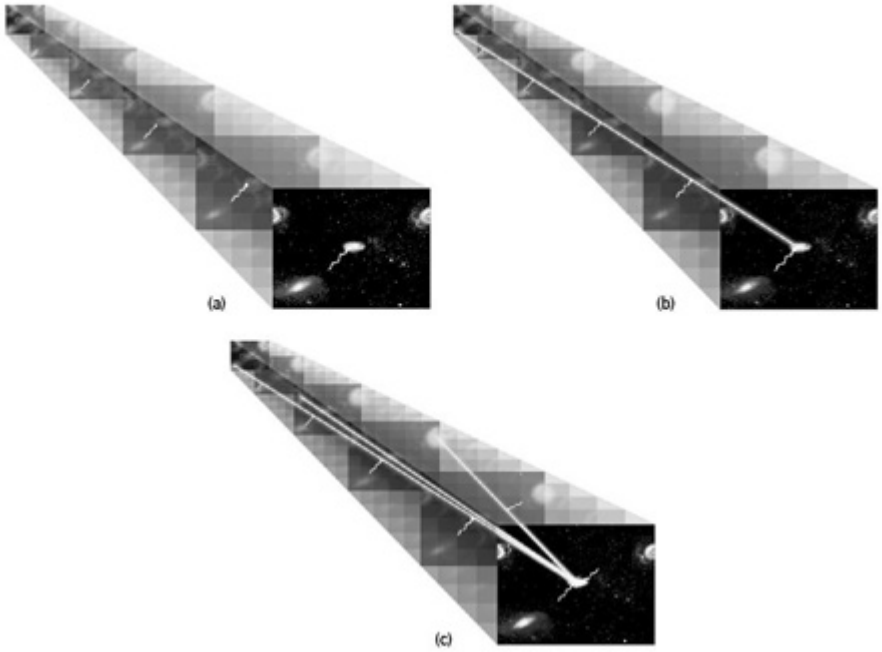


Figure 8.8 **(**a**)** Light emitted long ago from a distant galaxy gets closer and closer to the Milky Way on subsequent time slices. **(**b **)** When we finally see the distant galaxy, we are looking at it across both space and time, since the light we see was emitted long ago. The path through spacetime followed by the light is highlighted. **(**c**)** The paths through spacetime taken by light emitted from various astronomical bodies that we see today.

The figures dramatically show how light from space can be used as a cosmic time capsule. When we look at the Andromeda galaxy, the light we receive was emitted some 3 million years ago, so we are seeing Andromeda as it was in the distant past. When we look at the Coma cluster, the light we receive was emitted some 300 million years ago and hence we are seeing the Coma cluster as it was in an even earlier epoch. If right now all the stars in all the galaxies in this cluster were to go supernova, we would still see the same undisturbed image of the Coma cluster and would do so for another 300 million years; only then would light from the exploding stars have had enough time to reach us. Similarly, should an astronomer in the Coma cluster who is on our current now-slice turn a superpowerful telescope toward earth, she will see an abundance of ferns, arthropods, and early reptiles; she won’t see the Great Wall of China or the Eiffel Tower for almost another 300 million years. Of course, this astronomer, well trained in basic cosmology, realizes that she is seeing light emitted in earth’s distant past,